Specs

Aircraft

About Us
Net Weight (with propellers and RTK module) [1]
951 g

Max Takeoff Weight
1,050 g

Dimensions (Folded/Unfolded)
Folded (without propellers): 223×96.3×122.2 mm (Length×Width×Height)
Unfolded (without propellers): 347.5×283×139.6 mm (Length×Width×Height)

Diagonal Length
Diagonal: 380.1 mm

Max Ascent Speed
6 m/s (Normal Mode)
8 m/s (Sport Mode)

Max Descent Speed
6 m/s (Normal Mode)
6 m/s (Sport Mode)

Max Flight Speed (at sea level, no wind)
15 m/s (Normal Mode)
Flying forward: 21 m/s, flying sideways: 20 m/s, flying backwards: 19 m/s (Sport mode) [2]

Max Wind Speed Resistance
12 m/s [3]

Max Take-off Altitude Above Sea Level
6000 m (without a payload)

Max Flight Time (without wind)
43 minutes [4]

Max Hover Time (without wind)
37 minutes [4]

Max Flight Distance
32 km [6]

Max Pitch Angle
30° (Normal Mode)
35° (Sport Mode)

Max Angular Velocity
200°/s

GNSS
GPS + Galileo + BeiDou + GLONASS (GLONASS is supported only when RTK module is enabled)

Hovering Accuracy Range
Vertical: ±0.1 m (Vision Positioning enabled); ±0.5 m (GNSS Positioning enabled); ±0.1 m (D-RTK enabled)
Horizontal: ±0.3 m (Vision Positioning enabled); ±0.5 m (HD Positioning enabled); ±0.1 m (RTK enabled)

Operating Temperature
-10° to 40° C (14° to 104° F)

Internal Storage
N/A

Motor Model Number
2008

Propeller Model Number
9453F Enterprise Edition

Light Sensor
Built-in module

RGB Camera

Image Sensor
4/3 CMOS
Effective Pixels: 20 MP

Lens
FOV: 84°
Equivalent focal length: 24 mm
Aperture: f/2.8 to f/11
Focus: 1 m to ∞

ISO Range
100-6400

Shutter speed
Electronic shutter: 8-1/8000 s
Mechanical shutter: 8-1/2000 s

Max Image Size
5280×3956

Photo Shooting Mode
Single shot: 20 MP
Timelapse: 20 MP
JPEG: 0.7/1/2/3/5/7/10/15/20/30/60 s
JPEG + RAW: 3/5/7/10/15/20/30/60 s
Panorama: 20 MP (original material)

Video Resolution
H.264:
4K: 3840×2160@30fps
FHD: 1920×1080@30fps

Max Video Bitrate
4K: 130Mbps
FHD: 70Mbps

Supported File System
exFAT

Image Format
JPEG/DNG (RAW)

Video Format
MP4 (MPEG-4 AVC/H.264)

Multispectral Camera

Image Sensor
1/2.8-inch CMOS, effective pixels: 5 MP

Lens
FOV: 73.91° (61.2° x 48.10°)
Equivalent focal length: 25 mm
Aperture: f/2.0
Focus: Fixed Focus

Multispectral Camera Band
Green (G): 560 ± 16 nm;
Red (R): 650 ± 16 nm;
Red Edge (RE): 730 ± 16 nm;
Near infrared (NIR): 860 ± 26 nm;

Gain Range
1x-32x

Shutter Speed
Electronic Shutter: 1/30~1/12800 s

Max Image Size
2592×1944

Image Format
TIFF

Video Format
MP4 (MPEG-4 AVC/H.264)

Photo Shooting Mode
Single shot: 5 MP
Timelapse: 5 MP
TIFF: 2/3/5/7/10/15/20/30/60 s

Video Resolution
H.264
FHD: 1920 x 1080@30fps
Video content: NDVI/GNDVI/NDRE

Max Video Bitrate
Stream: 60 Mbps

Gimbal

Stabilization System
3-axis mechanical gimbal (tilt, roll, pan)

Mechanical Range
Tilt: -135° to 45°
Roll: -45° to 45°
Pan: -27° to 27°

Controllable Range
Tilt: -90° to 35°
Pan: Uncontrollable

Max Control Speed (tilt)
100°/s

Angular Vibration Range
±0.007°

Sensing System

Sensing System Type
Omnidirectional binocular vision system, with an infrared sensor at the bottom of the aircraft

Forward
Distance Measuring Range: 0.5 m to 20 m
Detection Range: 0.5 m to 200 m
Effective Obstacle Avoidance Speed: Flight Speed ≤15 m/s
FOV: Horizontal 90°, vertical 103°

Backward
Distance Measuring Range: 0.5 m to 16 m
Effective Obstacle Avoidance Speed: Flight speed ≤12 m/s
FOV: Horizontal 90°, vertical 103°

Lateral
Distance Measuring Range: 0.5 m to 25 m
Effective Obstacle Avoidance Speed: Flight speed ≤15 m/s
FOV: Horizontal 90°, vertical 85°

Upward
Distance Measuring Range: 0.2 m to 10 m
Effective Obstacle Avoidance Speed: Flight Speed ≤6 m/s
FOV: Front and rear 100°, left and right 90°

Downward
Distance Measuring Range: 0.3 m to 18 m
Effective Obstacle Avoidance Speed: Flight speed ≤6 m/s
FOV: Front and rear 130°, left and right 160°

Operating Environment
Front, Rear, Left, Right, Above: Surfaces with clear patterns and adequate lighting (> 15 lux, environments with normal indoor fluorescent light exposure)
Below: Surfaces with diffuse reflection material and a reflectivity of >20% (such as walls, trees, people, etc.); Adequate lighting (>15 lux, environments with normal indoor fluorescent light exposure)

Video Transmission

Video Transmission System
DJI O3 Image Transmission Industry Edition

Live View Quality
Remote Controller: 1080p/30fps

Operating Band [7]
2.400-2.4835 GHz
5.725-5.850 GHz

Max Effective Signal Distance (Unobstructed, No interference) [8]
FCC: 15 km
CE: 8 km
SRRC: 8 km
MIC: 8 km

Max Transmission Distance (Obstructed) [9]
Strong Interference (urban landscapes, residential areas, etc.): 1.5-3 km (FCC/CE/SRRC/MIC)
Medium Interference (suburban landscapes, city parks, etc.): 3-9 km (FCC), 3-6 km (CE/SRRC/MIC)
Weak Interference (remote fields, open farmland, etc.): 9-15 km (FCC), 6-8 km (CE/SRRC/MIC)

Max Download Speed
15 MB/s (with DJI RC Pro Industry Edition)

Latency (depending on environment and mobile device)
Approximately 200 milliseconds

Antennas
4 antennas, 2 transmitting and 4 receiving

Transmitter Power (EIRP)
2.4 GHz: <33 dBm (FCC), <20 dBm (CE/SRRC/MIC)
5.8 GHz: <33 dBm (FCC), <30 dBm (SRRC), <14 dBm (CE)

Other
Supports the DJI Cellular module

DJI RC Pro Enterprise Edition

Image Transmission System
DJI O3 Image Transmission Industry Edition

Max Effective Signal Distance (Unobstructed, No interference) [8]
FCC: 15 km
CE: 8 km
SRRC: 8 km
MIC: 8 km

Operating Band of Image Transmission [7]
2.400-2.4835 GHz
5.725-5.850 GHz

Antennas
4 antennas, 2 transmitting and 4 receiving

Operating Band of Image Transmission and Transmitter Power (EIRP)
2.4 GHz: <33 dBm (FCC); <20 dBm (CE/SRRC/MIC)
5.8 GHz: <33 dBm (FCC); <14 dBm (CE); <23 dBm (SRRC)

Wi-Fi Protocol
802.11 a/b/g/n/ac/ax
Support 2×2 MIMO Wi-Fi

Wi-Fi Operating Band [7]
2.400-2.4835 GHz
5.150-5.250 GHz
5.725-5.850 GHz

Wi-Fi Operating Band and Transmitter Power (EIRP)
2.4 GHz: <26 dBm (FCC); <20 dBm (CE/SRRC/MIC)
5.1 GHz: <26 dBm (FCC); <23 dBm (CE/SRRC/MIC)
5.8 GHz: <26 dBm (FCC/SRRC); <14 dBm (CE)

Bluetooth Protocol
Bluetooth 5.1

Bluetooth Operating Band
2.400-2.4835 GHz

Bluetooth Transmitter Power (EIRP)
<10 dBm

Screen Resolution
1920×1080

Screen Size
5.5 inches

Screen Frame Rate
60 fps

Screen Brightness
1,000 nits

Touch-Screen
10-point multi-touch

Battery
Lithium-ion battery (5000 mAh, 7.2 V)

Charging Type
100W Battery Charging Hub or USB charger with 12V or 15V specifications is recommended

Rated Power
12 Watts

Storage Capacity
Internal Memory (ROM): 64 GB
Supports microSD card usage to increase storage capacity

Charging Time
Approx. 1.5 hours (measured when only using the 100W Battery Charging Hub to charge the remote control, or when using a 15V USB charger)
Approximately 2 hours (measured using a 12V USB charger)
Approximately 2 hours and 50 minutes (measured using the 100W Battery Charging Hub to charge the aircraft and remote control at the same time)

Operating Time
Approx. 3 hours

Video Output Port
Mini-HDMI Port

Operating Temperature
-10° to 40° C (14° to 104°F)

Storage Temperature Range
<1 month: -30° to 60° C (-22° to 140° F)
One to three months: -30° to 45° C (-22° to 113° F)
Three to six months: -30° to 35° C (-22° to 95° F)
More than six months: -30° to 25° C (-22° to 77° F)

Charging Temperature
5° to 40° C (41° to 104° F)

Supported DJI Aircraft [10]
DJI Mavic 3E
DJI Mavic 3T
DJI Mavic 3M

GNSS
GPS + Galileo + GLONASS

Dimensions
Antenna is folded and no control sticks are installed:
183.27×137.41×47.6 mm (Length×Width×Height)
Antenna unfolded and control sticks are installed:
183.27×203.35×59.84 mm (Length×Width×Height)

Weight
Approx. 680 g

Model Number
RM510B

Storage

Supported microSD Cards
Aircraft:
Please use a memory card with a speed rating of V30 or higher, or use a memory card from the recommended list

Recommended microSD Cards
Remote Controller:
SanDisk Extreme PRO 64GB V30 A2 microSDXC
SanDisk High Endurance 64GB V30 microSDXC
SanDisk Extreme 128GB V30 A2 microSDXC
SanDisk Extreme 256GB V30 A2 microSDXC
SanDisk Extreme 512GB V30 A2 microSDXC
Lexar 667x 64GB V30 A2 microSDXC
Lexar High-Endurance 64GB V30 microSDXC
Lexar High-Endurance 128GB V30 microSDXC
Lexar 667x 256GB V30 A2 microSDXC
Lexar 512GB V30 A2 microSDXC
Samsung EVO Plus 64GB V30 microSDXC
Samsung EVO Plus 128GB V30 microSDXC
Samsung EVO Plus 256GB V30 microSDXC
Samsung EVO Plus 512GB V30 microSDXC
Kingston Canvas Go! Plus 128GB V30 A2 microSDXC
Kingston Canvas React Plus 128GB V90 A1 microSDXC

Aircraft:
SanDisk Extreme 32GB V30 A1 microSDHC
SanDisk Extreme PRO 32GB V30 A1 microSDHC
SanDisk Extreme 512GB V30 A2 microSDXC
Lexar 1066x 64GB V30 A2 microSDXC
Kingston Canvas Go! Plus 64GB V30 A2 microSDXC
Kingston Canvas React Plus 64GB V90 A1 microSDXC
Kingston Canvas Go! Plus 128GB V30 A2 microSDXC
Kingston Canvas React Plus 128GB V90 A1 microSDXC
Kingston Canvas React Plus 256GB V90 A2 microSDXC
Samsung PRO Plus 256GB V30 A2 microSDXC

Battery

Capacity
5000 mAh

Standard Voltage
15.4 V

Max Charging Voltage
17.6 V

Battery Type
LiPo 4S

Chemical System
Lithium Cobalt

Energy
77 watt-hours

Weight
335.5 g

Charging Temperature
5° to 40° C (41° to 104° F)

Battery Charger

Input
100V to 240V (AC), 50Hz to 60Hz, 2.5A

Output Power
100 Watts

Output
Maximum output power of 100 Watts (total)
When both the ports are used, the maximum output power of one interface is 82 W, and the charger will dynamically allocate the output power of the two interfaces according to load power.

Charging Hub

Input
USB-C: 5V to 20V, 5.0A

Output
Battery Port: 12V to 17.6V, 8.0A

Rated Power
100 Watts

Charging Type
3 batteries on charging rotation

Charging Temperature
5° to 40° C (41° to 104° F)

RTK Module

Dimensions
50.2×40.2×66.2 mm (Length×Width×Height)

Weight
24±2 g

Interface
USB-C

Power
Approximately 1.2 watts

RTK Position Accuracy
Fixed RTK:
Horizontal: 1 cm + 1 ppm; Vertical: 1.5 cm + 1 ppm

Notes

Footnotes

1. Standard weight of the aircraft (including battery, propellers, and microSD card). The actual product weight may vary due to differences in batch materials and external factors. Use for reference only.
2. The max speed in the EU cannot exceed 19 m/s.
3. Max wind resistance during takeoff and landing.
4. Data measured using the DJI Mavic 3M in a wind-free environment while flying at sea level at a constant speed of 36 kph until there was 0% power remaining. For reference only. Please pay attention to Return to Home prompts in the DJI Pilot 2 app when flying.
5. Data measured using the DJI Mavic 3M in a wind-free environment hovering over the sea level until there was 0% power remaining. For reference only. Please pay attention to Return to Home prompts in the DJI Pilot 2 app when flying.
6. Data measured using the DJI Mavic 3M in a wind-free environment while flying at sea level at 57.6 kph until there was 0% power remaining. For reference only. Please pay attention to Return to Home prompts in the DJI Pilot 2 app when flying.
7. In some countries, the 5.1/5.8GHz frequencies are prohibited, or the 5.1GHz frequency is only allowed for indoor use. Please refer to local laws and regulations before use.
8. Data measured flying in an unobstructed outdoor environment free of interference. It shows the farthest communication range for one-way, non-Return to Home flights under each standard. Please pay attention to Return to Home prompts in the DJI Pilot 2 app when flying.
9. Data measured in an unobstructed environment with typical interference under various standards. The actual flight distance may vary and is for reference only.
10. The DJI RC Plus will support more DJI aircraft in the future.

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